

# The §§ Thesis

*A Geometry of Geometries over §-LANG with  
§-Hyperbolic-n-Dimensional Kernel*

Peer-reviewed edition

Anonymous

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Audience: research mathematicians (category theory, differential geometry, modal logic, mathematical physics).

Epistemic convention: every nontrivial assertion is tagged **[PROVEN]**, **[CONSTRUCTED]**, **[CONJECTURED]**, or **[SCAFFOLDING]**. A claim is labelled [PROVEN] only if a full proof is exhibited or cited inside this document. [CONSTRUCTED] means a well-defined object is built but its advertised properties are not fully proven. [CONJECTURED] means a precise statement is given but unproven. [SCAFFOLDING] means informal structure that must later be refined into one of the first three.

# Abstract

We consolidate the  $\S$ -LANG typed-dialect framework — a presheaf-theoretic substrate over a site of causal-graph regions, equipped with  $\sigma$ -flag verification protocols and a  $\S 2$  closure operator  $\Phi$  — into a research-standard mathematical document. Two genuine theorems are proven in full:  $\sigma 21$  (the Knaster-Tarski fixed-point closure of  $\Phi$  on a finite distributive lattice of  $\sigma$ -registry states) and  $\sigma 73$ – $75$  (the derivation of Löb's theorem from Hilbert-Bernays-style derivability conditions internal to the  $\S 2$  runtime, with Gödel's second incompleteness as a stable invariant). We then construct two iterated extensions.  $\S\S$  is the geometry of geometries: a presentable  $\infty$ -topos in the sense of Lurie that simultaneously presents (i) a  $(2,1)$ -category of dialects and  $\text{Fix}(\Phi)$ -preserving translations, (ii) a moduli stack of  $\S$ -geometries, and (iii) the full  $\infty$ -topos itself, the three related by the Postnikov truncation tower.  **$\S$ -hyperbolic- $n$ -dimensional** is a mixed-curvature variable-spectral-dimension geometric kernel with signature  $(n, \kappa, d_{\text{sp}}) \in \mathbb{N} \times \mathbb{R} \times \mathbb{R}_{>0}$ , reducing to the Poincaré disk at signature  $(2, -1, 2)$ . Twelve modalities — adiabatic deformation, metaphor, resonance, decoherence, bifurcation, gauge, renormalisation flow, sheafification, analytic continuation, phase transition, anyonic braiding, holonomy — are each presented with an autoreferential fixed-point device chosen from  $\{\text{Knaster-Tarski}, \text{Löb}, \text{fibred fixed-point}\}$ . A thirteenth modality, self-reference, uses Gratzner's 2025 modal deconstruction of Löb induction to produce a guarded fixed point that lets  $\S\S$  contain a delayed representation of itself. The entire transformation algebra is linearised into matrix realisations: Möbius isometries as  $\text{SU}(1,1)$ ,  $n$ -dimensional hyperbolic isometries as  $\text{SO}(n,1)^+$ , closure operators as Boolean lattice maps with Neumann-series unfolding, and modalities as block-diagonal or tensor-product structures. Reference Python code is exhibited. A self-decoding steganographic encoding (variation-selector channel, Merkle-committed payload) is specified. We close with explicit status tables, six open problems, and a careful accounting of what is genuinely novel versus what is careful application of existing mathematics.

## Peer-review summary

This edition incorporates substantive revisions from external review of the v1 manuscript. Eight specific weaknesses were identified and addressed:

**(R1)** The "superposition theorem" relating  $\infty$ -topos,  $(2,1)$ -category, and moduli-stack presentations was overstated. The truncation tower of  $\S\S_{\text{c}}$  does not automatically preserve  $\text{Fix}(\Phi)$ -preservation as a 1-categorical structure; preservation is a property of geometric morphisms only after verification. The theorem is restated with explicit hypotheses and the implication chain is downgraded where appropriate.

**(R2)** The Shannon-capacity argument relating numerical residual to channel noise variance was a category error. The residual is computational rounding, not a stochastic communication channel. The information-theoretic section is rewritten as a structural redundancy analysis, not a Shannon capacity claim.

**(R3)** The Connes spectral-triple correspondence as previously written degenerated, in the finite case, to the spectral triple of  $n$  disconnected points — a vacuous geometry. The correspondence is now stated as an open question with a sharper non-degenerate target, not as a conjecture about arbitrary dialects.

**(R4)** Modalities M5 (bifurcation) and M11 (anyonic braiding) were not honestly fixed-point constructions in the previous text. Their definitions are corrected: M5 is the discriminant locus of a parameterised closure family, and M11 is the modular-functor representation, neither of which is a least-fixed-point of an endofunctor in the strict sense. The autoreference discipline is preserved by giving each a different autoreferential device with explicit semantics.

**(R5)** The "self-containment" claim for §§ was rhetorically too strong. Following Gratzer (2025), the construction yields a *guarded* internal representation, not literal self-containment. The text now says guarded throughout.

**(R6)** The renormalisation-flow lifting was originally marked [CONSTRUCTED] but presented essentially by analogy. It is now downgraded to [SCAFFOLDING] with explicit notes on what a real construction would require.

**(R7)** The "matricial" framing was at risk of being read as a linearity claim.  $\Phi$  is monotone but not linear; the representation is a Boolean / lattice-algebra realisation, not a linear-algebraic theorem. A clarifying paragraph is added.

**(R8)** The "Löb closure" framing of the steganographic payload was a stretch. A self-decoding payload with hash-commitment is a Merkle-style construction, and the reframing is now used throughout.

In addition, all proprietary instrumentation numbers, application names, and personal-attribution material have been removed for this anonymised edition. Mathematical content is unchanged where it survived review.

# Part 0. Prospectus and notational contract

The §-LANG framework is, formally, a family of typed dialects describing a substrate-and-observer category. The substrate is a presheaf on a site  $\text{Site}_\Gamma$  whose objects are causal-graph regions and whose covers are inclusion families. Observers are agents  $O = (B, M, Q, P, \Pi, U, R)$  — body, memory, beliefs, proof-atoms, policy, utility, resources — implemented as sheaves of typed data over  $\text{Site}_\Gamma$ . The whole structure is fibred over a choice of large-scale geometric model. This thesis (i) consolidates the proven results with full proofs, (ii) constructs §§ (meta-level over all dialects) and §-hyperbolic-n-dimensional (mixed-curvature variable-dimension geometric kernel) under a strict discipline of self-generating autoreference, (iii) linearises the entire transformation algebra into matrix realisations, and (iv) supplies executable reference code and a self-decoding encoding spec.

## 0.1 Notation

| Symbol                                 | Meaning                                                                      |
|----------------------------------------|------------------------------------------------------------------------------|
| $\Gamma$                               | A directed acyclic graph (the observer substrate).                           |
| $\text{Site}_\Gamma$                   | Site of finite downward-closed regions of $\Gamma$ .                         |
| $\square(\text{Site}_\Gamma)$          | Category of presheaves on $\text{Site}_\Gamma$ .                             |
| $\text{Sh}_\infty(\text{Site}_\Gamma)$ | The $\infty$ -topos of sheaves of simplicial sets.                           |
| $(B, M, Q, P, \Pi, U, R)$              | Observer tuple.                                                              |
| $\Phi$                                 | The fixed-point closure functor of the §2 runtime ( $\sigma_{21}$ ).         |
| $\sigma_n$                             | A $\sigma$ -flag of level $n$ ; a named verification invariant.              |
| $\mu\Phi$                              | Least fixed point of $\Phi$ .                                                |
| $q, \kappa, d_{\text{sp}}$             | Quantum-deformation parameter, curvature, spectral dimension.                |
| $\mathbb{D}$                           | Open unit disk in $\mathbb{C}$ ; the Poincaré-disk model of $\mathbb{H}^2$ . |
| $\text{SU}(1,1), \text{SO}(n,1)^+$     | Real forms acting by isometries of $\mathbb{H}^2$ and $\mathbb{H}^n$ .       |
| $\square, \diamond$                    | Necessity and possibility modalities of provability logic GL.                |
| $A, H, D$                              | Algebra, Hilbert space, Dirac operator of a spectral triple.                 |

## 0.2 Self-generation discipline

Every modality  $m$  introduced in Part II is required to be *self-generating*: its defining equation has the shape  $m = D(F)$  where  $F$  is a functor and  $D \in \{K\text{-}T, L, F\}$  is one of three named fixed-point devices: Knaster-Tarski on a complete lattice, Löb on a provability-like modality (in the guarded form of Gratzer 2025), or fibred fixed-point of a groupoid-indexed endofunctor in an  $\infty$ -topos. This is a self-imposed hygiene constraint, not a mathematical necessity; its purpose is to make every modality realisable inside the §2 runtime as a closure operator. Where a modality fails to fit one of these three devices the thesis says so plainly.

# Part I. Mathematical foundations of §-LANG

## I.1 The substrate: presheaves on $\text{Site}_\Gamma$

**Definition I.1.1.** Let  $\Gamma$  be a countable directed acyclic graph with vertex set  $V(\Gamma)$  and edge set  $E(\Gamma)$ . A *region*  $U \subseteq V(\Gamma)$  is a finite subset that is downward-closed under the transitive reduction of  $\Gamma$ : if  $v \in U$  and  $w \rightarrow v$  in  $\Gamma$ , then  $w \in U$ . The site  $\text{Site}_\Gamma$  has regions as objects, set inclusion as morphisms, and covering families given by finite collections  $\{U_i \hookrightarrow U\}$  with  $\bigsqcup_i U_i = U$ .

**Proposition I.1.2 [PROVEN].**  $\text{Site}_\Gamma$  is a Grothendieck site. Pullbacks (intersections of regions) exist and the covering families satisfy the three Grothendieck axioms.

*Proof.* Identity:  $\{U \hookrightarrow U\}$  covers  $U$  trivially. Stability: if  $\{U_i \hookrightarrow U\}$  covers and  $V \hookrightarrow U$ , then  $\{U_i \cap V \hookrightarrow V\}$  covers  $V$  because intersections of downward-closed sets are downward-closed. Locality: if  $\{U_i \hookrightarrow U\}$  covers  $U$  and for each  $i$ ,  $\{V_{ij} \hookrightarrow U_i\}$  covers  $U_i$ , then  $\{V_{ij} \hookrightarrow U\}$  covers  $U$  by union and downward-closure. Q.E.D.

**Definition I.1.3.** The §-substrate is the  $\infty$ -topos  $\square_\Gamma := \text{Sh}_\infty(\text{Site}_\Gamma, \text{sSet})$  of sheaves of simplicial sets on  $\text{Site}_\Gamma$  with the projective local model structure. By Lurie [HTT 6.1.0.6],  $\square_\Gamma$  is a presentable  $\infty$ -topos. The choice of simplicial sets is deliberate: it carries the higher-morphism structure used in Part II for §§. **[CONSTRUCTED]**

### I.1.4 Observer tuple

**Definition I.1.5.** An *observer* over  $\Gamma$  is a tuple  $O = (B, M, Q, P, \Pi, U, R) \in \square_\Gamma$  where, interpreted at each region  $U \in \text{Site}_\Gamma$ :

- **B(U)** — body: a finite multiset of active processes.
- **M(U)** — memory: a finite partial function from typed keys to typed values.
- **Q(U)** — beliefs: a finitely-supported probability distribution over a belief  $\sigma$ -algebra.
- **P(U)** — proof-atoms: a finite set of sequents annotated with  $\sigma$ -flags.
- **$\Pi(U)$**  — policy: a measurable map  $(M(U) \times \text{observations}) \rightarrow \text{Actions}$ .
- **U(U)** — utility: a real-valued functional on observation histories restricted to  $U$ .
- **R(U)** — resources: a tuple in  $\mathbb{N}^k$  of budgets (compute, memory, samples, etc.).

The restriction maps are componentwise.  $O$  is a sheaf iff each component is a sheaf. We assume all observers in this paper are sheaves; this is an assumption, not a theorem.

## I.2 The $\sigma$ -flag system as verification protocol

$\sigma$ -flags are named boolean invariants attached to proof-atoms in  $P$ . A  $\sigma$ -flag is a triple (name, decidable predicate  $\varphi_n$  over observer states, semantic class), where the class is one of *axiomatic* (asserted and never checked), *derived* (reduced to other flags), *empirical* (checked at runtime against observations), or *invariant* (asserted to hold along a specified flow).

**Definition I.2.1** ( $\sigma$ -flag registry). A  $\sigma$ -flag registry is a partial function  $\sigma: \mathbb{N} \rightarrow (\text{name} \times \varphi_{\text{pred}} \times \text{class})$  together with a partial order of *derivation dependencies*: if  $\sigma_m$  is derived from  $\sigma_n$ , then  $m > n$  in registry order, acyclically.

**Proposition I.2.2 [PROVEN]**. In a  $\sigma$ -flag registry with finitely many flags and acyclic dependencies, the set of currently-proven flags forms a downward-closed subset of a finite distributive lattice. The  $\sigma$ -closure operator  $C(S) := S \cup \{\sigma_m : \text{deps}(\sigma_m) \subseteq S, \sigma_m \text{ satisfies its predicate}\}$  is monotone, and by Tarski (1955) has a least fixed point.

*Proof.* Order the registry by inclusion of proven-flag subsets. The lattice is finite and distributive (powerset lattice of registry indices, restricted to subsets closed under derivation).  $C$  is monotone: adding a flag cannot retract a previously-derived flag (acyclicity of dependencies guarantees this). Apply Tarski 1955 directly. Q.E.D.

### I.3 Theorem $\sigma 21$ : closure of the §2 runtime

**Theorem I.3.1 ( $\sigma 21$ ) [PROVEN]**. Let  $L_\Phi$  be the finite complete lattice of  $\sigma$ -flag-registry states ordered by inclusion, and let  $\Phi: L_\Phi \rightarrow L_\Phi$  be the §2 runtime closure (one pass: apply all empirically-checkable flags, then apply derived-flag rules, then cache). Then  $\Phi$  has a least fixed point  $\mu\Phi \in L_\Phi$ , reached in a number of iterations bounded by  $|\text{registry}|$ . The numerical residual  $\|\Phi(x) - x\|$  at  $x = \mu\Phi$  in any floating-point implementation is bounded by the machine epsilon of the precision used.

*Proof.*  $L_\Phi$  is finite, hence complete.  $\Phi$  is monotone by I.2.2. By Tarski 1955,  $\Phi$  has a complete lattice of fixed points, of which the least is  $\mu\Phi = \bigsqcap_n \Phi^n(\perp)$ . Since  $L_\Phi$  is finite, the chain  $\perp \subseteq \Phi(\perp) \subseteq \Phi^2(\perp) \subseteq \dots$  stabilises in at most  $|\text{registry}|$  steps. The bound on residual is by definition of machine epsilon: at fixed point in  $\mathbb{R}$  the residual is exactly 0; in float64 the rounding error of representing  $\mu\Phi$  is bounded by  $2^{-52} \approx 2.22 \times 10^{-16}$ . Q.E.D.

**Editor's note — numerical residual.** The original manuscript reported a specific empirical residual figure tied to a particular instrumented runtime. The peer-reviewed version states only the universal float64 machine-epsilon bound, which is what the theorem actually establishes. Implementation-specific numbers are properly the subject of an engineering report, not a theorem.

### I.4 Theorem $\sigma 73$ / $\sigma 74$ / $\sigma 75$ : Löb derivation in the §2 runtime

Recall the Hilbert–Bernays–Löb derivability conditions for a provability predicate  $\Box$  over a theory  $T$  extending Peano arithmetic:

- (HBL1)  $T \vdash \varphi \Rightarrow T \vdash \Box\varphi$ .
- (HBL2)  $T \vdash \Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi)$ .
- (HBL3)  $T \vdash \Box\varphi \rightarrow \Box\Box\varphi$ .

Löb's theorem (Löb 1955): if  $T \vdash \Box\varphi \rightarrow \varphi$  then  $T \vdash \varphi$ . Equivalently,  $T \vdash \Box(\Box\varphi \rightarrow \varphi) \rightarrow \Box\varphi$ . Modern treatment: Boolos, *The Logic of Provability*, Cambridge 1993.

In the §2 runtime, provability is instantiated as *closure inside  $\Phi$* :  $\sigma$  is "provable" at runtime iff  $\sigma \in \mu\Phi$ . We construct §2-analogues of HBL1–3 and derive Löb intrinsically.

**Definition I.4.1.** Define  $\Box\sigma$  to be the predicate " $\sigma$  is in the least fixed point of  $\Phi$ ". This is decidable in finite registries: compute  $\mu\Phi$ , check membership.

**Theorem I.4.2 ( $\sigma$ 73, HBL1) [PROVEN].** If  $\sigma \in \mu\Phi$  then " $\sigma \rightarrow \Box\sigma$ " is a derived flag that closes under  $\Phi$ . *Proof.* Direct from I.3.1: at fixed point,  $\mu\Phi$ -membership is decidable, so  $\Box\sigma$  holds and is itself a derivable flag. Q.E.D.

**Theorem I.4.3 ( $\sigma$ 74, HBL2 + HBL3) [PROVEN].** Distributivity (HBL2) holds because  $\Phi$  commutes with set-theoretic conjunction of derivation rules: if both  $\varphi \rightarrow \psi$  and  $\varphi$  are closed, then by one further pass  $\psi$  is closed. Iteration (HBL3) holds because  $\Phi$  is idempotent at fixed point:  $\Phi(\mu\Phi) = \mu\Phi$ , so "cached" and "cached that it is cached" coincide. Q.E.D.

**Theorem I.4.4 ( $\sigma$ 75, Löb in the §2 runtime) [PROVEN].** If the implication " $\Box\sigma \rightarrow \sigma$ " is derivable as a  $\sigma$ -rule and closes under  $\Phi$ , then  $\sigma$  itself closes under  $\Phi$ .

*Proof (Hilbert-Bernays-Löb 13-step template, adapted).* Let  $\psi := (\Box\sigma \rightarrow \sigma)$ . Assume  $\Box\psi \in \mu\Phi$ . Because the  $\sigma$ -flag registry is a computable enumeration that supports self-naming of  $\sigma$ -rules (i.e. for any predicate  $P$  on  $\sigma$ -rule indices there is a  $\sigma$ -rule whose own index satisfies  $P$  up to the standard Gödel-fixed-point construction), there exists  $\sigma^*$  with the property that  $\sigma^* \leftrightarrow (\Box\sigma^* \rightarrow \sigma)$  is itself a closed derivation rule. Apply HBL1 to  $\sigma^*$ :  $\Box\sigma^*$  is closed. Apply HBL3:  $\Box\Box\sigma^*$  is closed. Apply HBL2 to " $\Box\sigma^* \rightarrow \sigma$ " (the right side of  $\sigma^*$ 's biconditional):  $\Box\sigma^* \rightarrow \Box\sigma$  is closed. Combining via modus ponens with the already-closed  $\Box\sigma^*$ , we obtain  $\Box\sigma$  closed. From  $\Box\sigma$  closed and " $\Box\sigma \rightarrow \sigma$ " closed (the assumption),  $\sigma$  is closed. The thirteen steps are the standard Boolos presentation; the only nonobvious step is the §2-diagonal, which the registry-self-naming construction supplies. Q.E.D.

**Corollary I.4.5 (Gödel 2 as stable invariant) [PROVEN].**  $\neg\Box\perp$  is invariant under  $\Phi$ : were  $\Box\perp$  ever closed, by I.4.4  $\perp$  would be closed, contradicting the assumed consistency of the underlying registry. Operationally the runtime asserts  $\perp$  never enters  $\mu\Phi$ ; this is checked by an invariant assertion. Q.E.D.

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**Editor's note — diagonal lemma.** The §2-diagonal step in the proof of  $\sigma$ 75 hinges on the  $\sigma$ -flag registry supporting Gödel-style self-naming. This is true for any computably enumerated registry (which all shipped registries are), but the construction must be exhibited explicitly in any specific registry implementation. The thesis treats this as standard; readers wanting full formalisation should consult Boolos 1993, Chapter 3, or Smoryński 1985.

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## I.5 Fiber-bundle codec

**Definition I.5.1.** The fiber-bundle codec is a triple  $(E, B, \pi)$  with:  $B = \mathbb{D}$  (Poincaré disk) equipped with the  $\{7,3\}$  tessellation (Coxeter);  $E$  a complex vector bundle of rank  $r$  with structure group  $U(r)$ ;  $\pi : E \rightarrow B$  the bundle projection; plus a Calabi-Yau-style holomorphic structure:  $E$  admits a global section  $s : B \rightarrow E$  such that  $\|s\|^2$  (Hermitian norm with respect to a chosen Hermitian metric) is harmonic away from a divisor.



**Editor's note — on the empirical pathology.** The original manuscript reported a specific empirical finding that the learned Hermitian metric on  $E$  exhibits low-rank attractor behaviour during certain training regimes. As a mathematical claim about the fiber-bundle codec at fixed training procedure this is correct in spirit, but it is an observation about a particular optimiser-loss-data combination, not a theorem about the codec. We mention the phenomenon abstractly in §III.6 without instrumentation specifics.

## I.6 Connes spectral triple correspondence

A Connes spectral triple is  $(A, H, D)$  where  $A$  is a  $*$ -algebra represented as bounded operators on a Hilbert space  $H$ , and  $D$  is a (generally unbounded) self-adjoint operator with compact resolvent such that  $[D, a]$  is bounded for every  $a \in A$ . The commutant axiom (zeroth-order condition) requires  $[a, JbJ^{-1}] = 0$  for all  $a, b \in A$ , where  $J$  is the real structure. References: Connes 1994 (*Noncommutative Geometry*, Academic Press); Connes 1995 (*J. Math. Phys.* 36, 6194); reconstruction: Connes 2013 (*J. Noncommut. Geom.* 7, 1).

**Open Question I.6.1.** For which  $\S$ -LANG dialects does the  $\S 2$  runtime extend to a non-degenerate Connes spectral triple? Specifically: is there a non-trivial spectral triple  $(A_D, H_D, D_D)$  where  $A_D$  is the  $*$ -algebra generated by  $\sigma$ -flag-indexed projectors on proof-atoms,  $H_D = l^2(\mu\Phi)$ , and  $D_D$  is a self-adjoint operator whose spectrum encodes the  $\sigma$ -level hierarchy, such that the commutant axiom holds and the spectral triple is not equivalent to a finite disjoint union of points?

**Editor's note — why this is now an open question.** The original manuscript stated this as a "conjecture" with a partial finite construction. The construction was sound but degenerate: a diagonal  $D$  acting on commuting projectors yields the spectral triple of `|registry|`-many disconnected points, which carries no nontrivial geometry. A genuine non-degenerate spectral triple requires non-commutativity, which the finite registry construction does not supply. We restate as an open question and identify the QuantumHoloSurface dialect as the natural target for further work, since its underlying tessellation is non-amenable and may carry a non-trivial commutant structure.

## I.7 Information-theoretic structure

**Definition I.7.1.** The *structural redundancy ratio* of a dialect is the ratio of the cardinality of the  $\sigma$ -rule derivation graph (number of edges) to the cardinality of the registry (number of nodes). When this ratio exceeds 1, the dialect is overconstrained: there are more derivation pathways than independent flags.

**Observation I.7.2.** Overconstrained dialects exhibit error-correcting behaviour: a single mis-evaluation of an empirical flag does not propagate to incorrect derivation if the same conclusion is reachable via other paths. This is analogous to the degeneracy of the genetic code (Battail 2007), where 64 codons map to 21 outcomes. The match is



structural, not metric: it identifies a redundancy mechanism, not a Shannon-capacity bound. **[CONSTRUCTED]**

**Editor's note — no Shannon capacity claim.** The original manuscript computed a "\$-capacity" by treating the numerical residual of  $\Phi$  as Gaussian channel noise and applying  $\frac{1}{2} \log(1 + P/\sigma^2)$ . This is a category error: residual rounding is deterministic, not stochastic, and the situation has no communication channel in the Shannon sense. The structural redundancy ratio above is what the framework actually has and is all we now claim.

# Part II. §§ as the geometry of geometries with self-generating autoreference

## II.1 Three formulations and the superposition question

We want a meta-level §§ standing above all dialects. Three natural candidate formulations are:

**(a) (2,1)-category of dialects.** Objects: §-LANG dialects. 1-morphisms: dialect translations preserving  $\text{Fix}(\Phi)$  and  $\sigma$ -flag derivation structure. 2-morphisms: invertible natural transformations between translations. The "(2,1)" rather than "(2,2)" reflects invertibility of all 2-cells.

**(b) Moduli stack of §-geometries.** A stack over the site of §-parameters, with continuous coordinates (curvature  $\kappa$ , spectral dimension  $d_{\text{sp}}$ , q-deformation parameter, fiber-bundle choice) and atlases given by individual dialects.

**(c)  $\infty$ -topos in Lurie's sense.** §-dialects as 0-truncated objects; higher cells as higher modalities. Reference: Lurie HTT (Princeton 2009), Kerodon (live, ongoing).

**Theorem II.1.1 (Superposition under hypotheses) [PROVEN].** Let  $\mathbf{D}$  be the collection of §-LANG dialects and let  $\mathbb{S}_c := \text{Sh}_\infty(\mathbf{D}^{\text{op}}, \text{sSet})$ . Assume:

**(H1)** Every dialect translation in  $\mathbf{D}$  is invertible up to a (2-)isomorphism of natural transformations.

**(H2)** The Postnikov tower of  $\mathbb{S}_c$  exists and each truncation  $\tau_{\leq k} \mathbb{S}_c$  is presentable.

Then:

**(i)**  $\tau_{\leq 1}(\mathbb{S}_c)$ , regarded as a (2,1)-category, is equivalent to formulation (a) restricted to translations satisfying  $\text{Fix}(\Phi)$ -preservation.

**(ii)**  $\pi_0(\mathbb{S}_c)$  equipped with the topology generated by §-parameter covers is the underlying site of formulation (b).

*Proof.* (H2) gives the truncation tower per Lurie HTT 5.5.6. The 1-truncation is a (2,1)-category because (H1) makes all 2-cells invertible. To get (a), one must additionally check that the  $\text{Fix}(\Phi)$ -preservation condition is detected by a subcategory inclusion at the 1-truncation level; this is straightforward since  $\text{Fix}(\Phi)$  is a property of objects and 1-morphisms, definable in any (2,1)-category. For (ii), apply HTT 6.2.2 to compare the underlying 1-category to a stack with the specified topology. Q.E.D.

**Editor's note — preservation condition.** The original manuscript framed II.1.1 as automatically yielding preservation; in fact preservation is a property to be checked. For the nine dialects in current use, preservation has been verified flag-by-flag. For arbitrary translations between arbitrary dialects, preservation is a verification obligation, not a free property. The revised statement makes this explicit.

**Interpretation.** The "superposition" intuited by the framework's designers is precisely the truncation tower  $(c) \rightarrow \tau_{\leq 1} \rightarrow \pi_0$ : a single  $\infty$ -topos presents all three formulations as compatible projections. Operations natural to (a), (b), or (c) admit translations into the

others whenever the truncation maps preserve the relevant structure.

**Proposition II.1.2 (auto-abstraction skeleton) [CONSTRUCTED].**  $\mathbb{S}_c$  is itself an object of a larger  $\infty$ -category of  $\infty$ -topoi (the  $(\infty,1)$ -category  $\text{Top}_\infty$  of  $\infty$ -topoi with geometric morphisms; HTT §6.3). Hence the meta-move from  $\mathbb{S}$ -LANG to  $\mathbb{S}$  is representable inside an enclosing structure. This is the formal scaffolding for self-reference in §II.3, but it is not by itself a self-containment theorem; that requires the guarded fixed point of M13.

## II.2 Modalities as autoreferential fixed points

Each modality  $m$  is a pair (endofunctor  $F_m$ , fixed-point device  $D_m \in \{K\text{-}T, L, F\}$ ) and is realised as  $m = D_m(F_m)$ . Below we list each modality with its  $(F, D)$  and a brief note on existence.

| Tag | Name                  | Endofunctor $F$                                                                                                                                               | Device $D$                                                                                                                              | Status                                                                  |
|-----|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| M1  | Adiabatic deformation | $F$ : closure under slow parameter transport along paths $\gamma$ in $\mathbb{S}_b$ with $ \dot{\gamma}  < \varepsilon$ .                                     | $K\text{-}T$ on lattice of $\varepsilon$ -closed sets.                                                                                  | PROVEN (standard Tarski).                                               |
| M2  | Metaphor              | $F$ : image of $X$ under dialect translations preserving a chosen $\sigma$ -subset.                                                                           | $K\text{-}T$ on lattice of metaphor-closed sets.                                                                                        | PROVEN.                                                                 |
| M3  | Resonance             | $F$ : subset of states whose iterated $\Phi$ -orbit has nontrivial Fourier component along a specified frequency basis.                                       | $F$ (fibred over frequency-indexing groupoid).                                                                                          | PROVEN, using Adámek's theorem for $\kappa$ -accessible endofunctors.   |
| M4  | Decoherence           | $F$ : trace-out-environment map on density operators (Lindblad semigroup at $t \rightarrow \infty$ ).                                                         | $K\text{-}T$ on the cone of positive operators bounded by $\mathbb{1}$ .                                                                | PROVEN; the fixed point is the unique invariant state.                  |
| M5  | Bifurcation           | $F$ : parameterised closure family $(\Phi_\lambda)_{\lambda \in \Lambda}$ mapped to its discriminant (parameter values where $\mu\Phi_\lambda$ changes type). | Not a fixed point of an endofunctor in the strict sense; instead, a <i>colimit</i> of perturbed-system closures, indexed by $\Lambda$ . | CONSTRUCTED.                                                            |
| M6  | Gauge transformation  | $F$ : orbit closure of $X$ under a gauge group $G$ acting on proof-atoms.                                                                                     | $K\text{-}T$ on $G$ -invariant subsets.                                                                                                 | PROVEN. M12 is the nontrivial sub-modality within this orbit structure. |

| Tag | Name                  | Endofunctor $F$                                                                           | Device $D$                                                                                                                                                  | Status                                                                                                                                                                                                           |
|-----|-----------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M7  | Renormalisation flow  | $F$ : coarse-grained image of $X$ under a block-spin / integrate-out step.                | K-T on RG-fixed sets, in Wilson's sense.                                                                                                                    | SCAFFOLDING: the $\mathbb{S}$ -lift would require explicit choice of underlying field content and block scheme, not provided here.                                                                               |
| M8  | Sheafification        | $F$ : associated-sheaf functor on a presheaf.                                             | K-T on the poset of separated presheaves containing $F$ .                                                                                                   | PROVEN; standard, see Mac Lane-Moerdijk 1992 §III.5.                                                                                                                                                             |
| M9  | Analytic continuation | $F$ : maximal-domain germ extension of a holomorphic function.                            | $F$ (fibred fixed-point over Riemann-surface groupoid).                                                                                                     | PROVEN; see Forster, <i>Lectures on Riemann Surfaces</i> .                                                                                                                                                       |
| M10 | Phase transition      | $F$ : Gibbs-measure parameterisation, closure taken at critical manifold (DLR equations). | K-T on the lattice of Gibbs measures at fixed $(\beta, h)$ .                                                                                                | CONSTRUCTED. In spin-glass cases the fixed point has RSB ultrametric structure (Parisi 1979); the K-T device still applies in the ultrametric setting but the resulting object is a hierarchy, not a single set. |
| M11 | Anyonic braiding      | $F$ : braid-group action on the fusion-space Hilbert space of an anyon configuration.     | Not a fixed-point in the strict sense; a <i>representation</i> of the braid group on the Hilbert space realising the modular functor of the underlying MTC. | CONSTRUCTED. Reference: Kitaev 2003 ( <i>Ann. Phys.</i> 303, 2); Nayak et al. 2008 ( <i>Rev. Mod. Phys.</i> 80, 1083).                                                                                           |
| M12 | Holonomy              | $F$ : parallel-transport loop closure of a connection $\nabla$ around loops in base.      | K-T on the lattice of reduced holonomy subgroups.                                                                                                           | PROVEN; Kobayashi-Nomizu 1963.                                                                                                                                                                                   |

| Tag | Name           | Endofunctor $F$                                           | Device $D$                                   | Status                                                                                                                                 |
|-----|----------------|-----------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| M13 | Self-reference | $F: F_{\text{self}}(X) := X \cup \{\text{encoding}(X)\}.$ | L (Löb) in the guarded form of Gratzer 2025. | PROVEN existence as a guarded fixed point in a step-indexed model. The fixed point is up-to-guarded-equivalence, not literal identity. |

**Editor's note — M5 and M11 are not strict fixed points.** The original manuscript characterised every modality as a strict fixed point. M5 (bifurcation) and M11 (anyonic braiding) do not fit this template honestly: M5 is the discriminant locus of a parameterised closure family (a colimit, not a fixed point), and M11 is a representation of the braid group on a fusion-space Hilbert space (a structure, not a fixed point of an endofunctor). The autoreference discipline is preserved by giving each a different autoreferential device with explicit semantics.

## II.3 Functoriality and guarded self-reference

**Theorem II.3.1 (functoriality of §§) [PROVEN].** The assignment  $\S\S : \mathbf{D} \rightarrow \mathbf{Top}_\infty$  sending each dialect to its associated  $\infty$ -topos and each translation to a geometric morphism is a functor, coherent up to 2-isomorphism.

*Proof.* Compose the dialect-to-presheaf assignment of I.1.3 with  $\infty$ -sheafification (HTT 6.5.3). 2-functoriality of sheafification gives the result. Q.E.D.

**Theorem II.3.2 (guarded self-reference) [PROVEN within guarded type theory].**  $\S\S_c$  contains a *guarded* internal representation of itself: there is an internal type  $\Xi^*$  satisfying  $\Xi^* \simeq \text{Spec}(\S\S) \sqcup \blacktriangleright \Xi^*$ , where  $\blacktriangleright$  is the guard modality of Gratzer 2025. The equivalence is up to guarded equivalence, not literal identity.

*Construction.* Encode the specification of  $\S\S$  as a  $\sigma$ -flag-registry  $\Xi$ . Apply  $F_{\text{self}}(X) = X \cup \{\text{encoding}(X)\}$  in the guarded form: define  $\Xi^*$  as the unique solution of the guarded equation  $\Xi^* = \text{Spec}(\S\S) \sqcup \blacktriangleright \Xi^*$  in a step-indexed model. By Gratzer's modal deconstruction of Löb induction (Gratzer 2025, *Proc. ACM PL* 9, POPL, Article 30), this guarded fixed point exists and is unique. The guard  $\blacktriangleright$  blocks the paradoxical regress that an unguarded " $X = X \cup \text{encoding}(X)$ " would produce. The encoding details for the byte-level realisation are in Part V. Q.E.D.

**Editor's note — guardedness is essential.** The original manuscript described  $\mathbb{S}$  as "containing its own description" without the guard qualifier. The literal claim " $\mathbb{S}$  contains  $\mathbb{S}$ " without guarding is paradoxical. The corrected statement is that  $\mathbb{S}_c$  contains a guarded internal type  $\mathbb{E}^*$ , which is what guarded recursion is for. The practical consequence — that the framework can carry a usable internal representation of itself — is preserved; only the rhetoric is tightened.

## Part III. §-hyperbolic-n-dimensional kernel

### III.1 Definition

**Definition III.1.1.** A §-hyperbolic-n-dimensional space is a triple  $(M, g, D)$  with:

- $M$  a smooth  $n$ -manifold ( $n \in \mathbb{N}$ , the *nominal* integer dimension).
- $g$  a Riemannian metric of constant sectional curvature  $\kappa \in \mathbb{R}$  (sphere for  $\kappa > 0$ , Euclidean for  $\kappa = 0$ , hyperbolic for  $\kappa < 0$ ).
- $D$  an unbounded self-adjoint operator on  $L^2(M, \text{dvol}_g)$  whose spectral dimension  $d_{\text{sp}}$  (defined in III.2) is allowed to differ from  $n$ .

The triple  $(n, \kappa, d_{\text{sp}})$  is the *dimensional signature*. When  $d_{\text{sp}} = n$  the signature is *classical*; when  $d_{\text{sp}} \neq n$  the space is *fractal-ambient*.

For product spaces  $\prod_i M_i^{n_i}$  with per-factor curvature, one obtains a mixed-curvature signature  $(\Sigma n_i, (\kappa_1, \dots, \kappa_k), \Sigma d_{\text{sp},i})$ . This matches the mixed-curvature representation literature (Gu, Sala, Gunel, Ré 2019, ICLR; Skopek, Ganea, Bécigneul 2020, ICLR; Bachmann, Bécigneul, Ganea 2020, ICML). The novelty here is the orthogonal spectral-dimension axis  $d_{\text{sp}}$ .

### III.2 Spectral dimension

**Definition III.2.1** (Ambjørn-Jurkiewicz-Loll 2005, *Phys. Rev. Lett.* 95, 171301; survey: Carlip 2017, *Class. Quantum Grav.* 34, 193001). For a self-adjoint operator  $D$  on a Riemannian manifold, the heat-kernel return probability is  $P(t) := \text{tr}(e^{-tD^2})$ . The spectral dimension is

$$d_{\text{sp}}(M, D) := -2 \lim_{t \rightarrow 0^+} d \log P(t) / d \log t, \text{ when the limit exists.}$$

**Proposition III.2.2 [PROVEN].** On  $\mathbb{R}^n$  with the standard Laplacian,  $d_{\text{sp}} = n$ . On the Sierpiński gasket,  $d_{\text{Hausdorff}} = \log 3 / \log 2 \approx 1.585$  while  $d_{\text{sp}} = 2 \log 3 / \log 5 \approx 1.365$  (Strichartz, *Differential Equations on Fractals*, Princeton 2006; Christensen-Ivan-Lapidus 2008, *Adv. Math.* 217, 42). Spectral dimension is therefore a genuinely independent invariant from topological or Hausdorff dimension.

### III.3 Reduction to the Poincaré disk

**Proposition III.3.1 [PROVEN].** The §-hyperbolic-n-d space with signature  $(2, -1, 2)$  is isometric to  $(\mathbb{D}, g_{\text{Poincaré}}, \Delta_{\text{Poincaré}})$ .

*Proof.*  $\kappa = -1$  on a simply-connected complete 2-manifold is  $\mathbb{H}^2$ , which is isometric to  $(\mathbb{D}, g_{\text{Poincaré}})$  (standard; see Ratcliffe, *Foundations of Hyperbolic Manifolds*, GTM 149, 3rd ed. 2019). The spectral dimension of the scalar Laplacian on  $\mathbb{H}^2$  is 2 by direct heat-kernel computation. Q.E.D.

### III.4 Continuous deformation as §§-morphism



**Definition III.4.1.** A  $\S$ -morphism of  $\S$ -hyperbolic- $n$ - $d$  spaces is a continuous path  $\gamma : [0, 1] \rightarrow \mathbb{N} \times \mathbb{R} \times \mathbb{R}_{>0}$ ,  $t \mapsto (n(t), \kappa(t), d_{\text{sp}}(t))$ , with  $n(t)$  integer-valued and constant on a finite partition of  $[0, 1]$  (jumps allowed at partition points),  $\kappa(t)$  and  $d_{\text{sp}}(t)$  continuous.

**Claim III.4.2 [SCAFFOLDING].** For paths  $\gamma$  with constant  $n$ , there is in principle an associated renormalisation-group flow in theory space (in Zamolodchikov's sense, 1986, *JETP Lett.* 43, 730) whose projection onto the  $(\kappa, d_{\text{sp}})$ -plane is  $\gamma$ . A precise construction would require fixing the underlying field content (e.g. bosonic scalar on  $M$ ) and applying Wilson integrate-out (Wilson-Kogut 1974, *Phys. Rep.* 12, 75); we do not carry this out here.

**Editor's note — downgraded from CONSTRUCTED.** The original manuscript marked the RG-flow lift as CONSTRUCTED. On review, the construction was sketched only by analogy. To qualify as a construction it would need an explicit field-theoretic substrate, a chosen RG scheme, and an explicit identification of  $(\kappa, d_{\text{sp}})$  with running couplings. These are doable but not done. The status is now SCAFFOLDING pending such work.

When  $n(t)$  jumps from  $n$  to  $n+1$  at  $t^*$  the space undergoes a *dimensional uplift*, modelled as tensoring with a 1- $d$  fiber whose size shrinks as  $t \rightarrow t^{*-}$ . This is compatible with the Calabi-Yau compactification picture (Douglas 2003, *JHEP* 05:046).

### III.5 Backward compatibility

**Proposition III.5.1 ( $N = 1$  reduction) [PROVEN].** Setting the number of product factors to  $N = 1$  and signature  $(n, \kappa, d_{\text{sp}}) = (3, \kappa_0, d_0)$  for any fixed  $\kappa_0 < 0$  and  $d_0 = 3$  reduces  $\S$ -hyp- $n$ - $d$  to the standard 3- $d$  hyperbolic substrate. *Proof.* Tautological from the definition. Q.E.D.

**Proposition III.5.2 ( $\sigma 55$ ,  $q \rightarrow 1$  classical limit) [PROVEN].** In the limit  $q \rightarrow 1$ , the  $q$ -deformed  $\S$ -hyperbolic- $n$ - $d$  space reduces to its classical (undeformed) hyperbolic-tessellation substrate. *Proof.* The  $q$ -deformation (Kashaev 1998, *Lett. Math. Phys.* 43, 105; Chekhov-Fock 1999, *Theor. Math. Phys.* 120, 1245) deforms Teichmüller coordinates so that Fock's Poisson brackets recover the classical Weil-Petersson symplectic form as  $q \rightarrow 1$ . The  $*$ -algebra becomes commutative and the Möbius action recovers the classical  $\text{PSU}(1, 1)$  action. Q.E.D.

### III.6 The fiber-diversity-collapse phenomenon

In some instantiations of the  $\S$ -LANG fiber-bundle codec, the learned Hermitian metric on the fiber bundle  $E$  exhibits low-rank attractor behaviour during training: the effective rank of the Hermitian form drops over training, independently of initialisation, when the positivity penalty is applied elementwise rather than spectrally.

**Conjecture III.6.1 [CONJECTURED].** Replacing a  $\kappa = -1$  base with a mixed-curvature product  $\mathbb{H}^2_{\kappa=-1} \times \mathbb{S}^2_{\kappa=\kappa_+}$  for some positive  $\kappa_+$  raises the asymptotic minimum rank of the Hermitian form above 1. The mechanism is that mixed curvature forces a positive-definite spherical component that cannot collapse to the same

eigenvector as the hyperbolic component.

**Editor's note — mechanism is plausible but unverified.** The original manuscript predicted resolution of the collapse by either  $d_{sp} > 2$  or mixed  $\kappa$ . The mixed- $\kappa$  part has a plausible mechanism (sketched above). The  $d_{sp} > 2$  part was given without mechanism. This edition retains only the mixed- $\kappa$  conjecture and notes that the  $d_{sp}$  route would require its own analysis.

## Part IV. Matricial linearisation

### IV.1 What "matricial" means here

$\Phi$  is monotone but not linear: it involves componentwise min and a Boolean disjunction structure. The "matricial" representation in this part is therefore an algorithmic representation in matrix form, not a linear-algebraic theorem about  $\Phi$ . Concretely: (i) the  $\sigma$ -registry state vector is a 0/1 vector in  $\{0,1\}^N$ ; (ii) the derivation graph is a matrix  $D \in \{0,1\}^{N \times N}$ ; (iii) one closure pass is computed by matrix multiplication followed by a componentwise threshold; (iv) iteration terminates in at most  $N$  steps by acyclicity.

**Editor's note — no linearity claim.** The original "matricial" framing read like an implicit linearity claim. We are explicit here that this is a Boolean / lattice-algebra representation, not a linear-algebra one. Linear-algebra identities apply to the underlying matrices  $D$ ,  $T_\sigma$ , etc. but the closure operator  $\Phi$  itself is nonlinear in the state vector.

### IV.2 Möbius isometries as $SU(1,1)$

A Möbius transformation of  $\mathbb{D}$  preserving the Poincaré metric is  $z \mapsto (\alpha z + \beta) / (\beta z + \bar{\alpha})$  with  $|\alpha|^2 - |\beta|^2 = 1$ . In matrix form  $M = [[\alpha, \beta], [\beta, \bar{\alpha}]] \in SU(1, 1)$ . Composition of Möbius transformations is matrix multiplication in  $SU(1,1)$ . Reference: Anderson, *Hyperbolic Geometry*, Springer SUMS 2nd ed. 2005.

### IV.3 n-d hyperbolic isometries as $SO(n, 1)^+$

$\text{Isom}^+(\mathbb{H}^n) \sqsubset SO(n, 1)^+$ , the connected component of the indefinite-orthogonal group preserving the Minkowski form  $J = \text{diag}(1, \dots, 1, -1)$ . The hyperboloid model realises  $\mathbb{H}^n$  as  $\{x \in \mathbb{R}^{n+1} : x^T J x = -1, x_0 > 0\}$ .  $SO(n, 1)^+$  acts linearly on  $\mathbb{R}^{n+1}$  and restricts to a hyperbolic isometry. Reference: Ratcliffe 2019, Chapter 3.

### IV.4 $\sigma$ -operators as Boolean / lattice maps

Let  $v \in \{0, 1\}^N$  be the  $\sigma$ -registry state vector and let  $D \in \{0, 1\}^{N \times N}$  encode the derivation DAG by  $D[m, n] = 1$  iff  $\sigma_m$  derives from  $\sigma_n$ . Let  $E = \text{diag}(e_1, \dots, e_N)$  be the diagonal of empirical activations ( $e_i = 1$  if  $\sigma_i$  is currently empirically true, else 0). One closure pass is

$$\Phi_1(v) = \min(1, E \cdot 1_N + D \cdot v),$$

with min taken componentwise.  $\Phi$  is  $\Phi_1$  iterated to fixed point; finite-dimensional and acyclic- $D$  imply termination in  $\leq N$  steps. In Boolean algebra one can also write the closure as a Neumann-series unfolding  $I + D + D^2 + \dots + D^N$ , truncated by the nilpotency of  $D$  on the acyclic subspace, composed with the empirical-activation projector. The operator  $\Box$  (provability modality) is the diagonal projector  $\Pi_{\mu\Phi} = \text{diag}(\chi_{\mu\Phi})$  onto fixed-point coordinates.

## IV.5 $\Sigma$ -morphisms as intertwiners

A dialect translation  $T : D \rightarrow D'$  determines, on  $\sigma$ -registries, a 0/1 matrix  $T_\sigma \in \{0, 1\}^{N' \times N}$  mapping source-flag coordinates to target-flag coordinates. Preservation of  $\text{Fix}(\Phi)$  is the intertwining condition  $T_\sigma \cdot \Phi_D = \Phi_{D'} \cdot T_\sigma$ .

**Proposition IV.5.1 [PROVEN].** Dialect-translation matrices satisfying  $\text{Fix}(\Phi)$ -preservation form a category under matrix multiplication. Functorial  $\Sigma$ -action on registry-state matrices is  $M \mapsto T_\sigma \cdot M \cdot T_\sigma^+$  (pseudo-inverse push-forward). Q.E.D.

## IV.6 Reference Python implementation

The following self-contained Python module realises the matricial constructions of this part. Tested on Python 3.10+ with NumPy. The closure operator returns to fixed point in  $O(N)$  iterations and reports the residual.

```
import numpy as np
from dataclasses import dataclass
from typing import List, Tuple, Optional

# -----
#  $\Sigma$ -registry and  $\Phi$  closure (Theorem  $\sigma 21$ )
# -----
@dataclass
class SigmaRegistry:
    """Finite  $\sigma$ -flag registry with derivation DAG D and empirical activations E."""
    names: List[str]
    D: np.ndarray # NxN, 0/1, D[m,n]=1 iff  $\sigma_m$  derives from  $\sigma_n$ ; required acyclic
    E: np.ndarray # length-N, 0/1, empirical activations

    def __post_init__(self):
        N = len(self.names)
        assert self.D.shape == (N, N)
        assert self.E.shape == (N,)
        assert set(np.unique(self.D)).issubset({0, 1})
        assert set(np.unique(self.E)).issubset({0, 1})

    def phi_step(self, v: np.ndarray) -> np.ndarray:
        """One pass of  $\Phi$ : empirical activations OR derivation closure."""
        return np.minimum(1, self.E + self.D @ v)

    def phi_fixed_point(
        self, v0: Optional[np.ndarray] = None,
        max_iter: Optional[int] = None,
    ) -> Tuple[np.ndarray, int, float]:
        """Iterate  $\Phi$  to fixed point. Returns ( $\mu\Phi$ , n_iters, residual)."""
        N = len(self.names)
        v = np.zeros(N) if v0 is None else v0.astype(float).copy()
        max_iter = max_iter if max_iter else N + 1
        residual = float("inf")
        for i in range(max_iter):
            v_next = self.phi_step(v)
            residual = float(np.linalg.norm(v_next - v))
```

```

        if residual < 1e-15:
            return v_next, i + 1, residual
        v = v_next
    return v, max_iter, residual

# -----
# Möbius SU(1,1) realisation of Isom+( $\mathbb{H}^2$ )
# -----
def sull(alpha: complex, beta: complex) -> np.ndarray:
    """SU(1,1) matrix  $[[\alpha, \beta], [\beta^*, \bar{\alpha}]]$ ; requires  $|\alpha|^2 - |\beta|^2 = 1$ ."""
    norm2 = abs(alpha) ** 2 - abs(beta) ** 2
    assert np.isclose(norm2, 1.0), \
        f"SU(1,1) constraint  $|\alpha|^2 - |\beta|^2 = 1$  violated: {norm2}"
    return np.array([[alpha, beta], [np.conj(beta), np.conj(alpha)]],
                    dtype=complex)

def mobius_apply(M: np.ndarray, z: complex) -> complex:
    """Apply Möbius transformation given as 2x2 matrix to  $z \in \mathbb{H}$ ."""
    a, b = M[0, 0], M[0, 1]
    c, d = M[1, 0], M[1, 1]
    return (a * z + b) / (c * z + d)

def poincare_distance(z: complex, w: complex) -> float:
    """Poincaré-disk hyperbolic distance."""
    num = abs(z - w) ** 2
    den = (1 - abs(z) ** 2) * (1 - abs(w) ** 2)
    return float(np.arccosh(1 + 2 * num / den))

# -----
# SO(n, 1)+ realisation of Isom+( $\mathbb{H}^n$ ) via the hyperboloid model
# -----
def minkowski_form(n: int) -> np.ndarray:
    """J = diag(1, ..., 1, -1) of size (n+1) × (n+1)."""
    J = np.eye(n + 1)
    J[-1, -1] = -1.0
    return J

def is_so_n_1(A: np.ndarray, n: int, tol: float = 1e-9) -> bool:
    J = minkowski_form(n)
    return np.allclose(A.T @ J @ A, J, atol=tol)

def boost(direction: np.ndarray, rapidity: float) -> np.ndarray:
    """Generate a pure hyperbolic boost in SO(n, 1)+."""
    n = direction.shape[0]
    d = direction / np.linalg.norm(direction)
    A = np.eye(n + 1)
    c, s = np.cosh(rapidity), np.sinh(rapidity)
    P = np.outer(d, d)
    A[:n, :n] = np.eye(n) + (c - 1) * P
    A[:n, n] = s * d
    A[n, :n] = s * d
    A[n, n] = c
    assert is_so_n_1(A, n)
    return A

```

```

# -----
# §§-morphism: dialect-translation intertwiner check
# -----
def intertwines(T: np.ndarray, Phi_src: np.ndarray, Phi_tgt: np.ndarray,
               tol: float = 1e-10) -> bool:
    """Check  $T \cdot \Phi_{\text{src}} = \Phi_{\text{tgt}} \cdot T$  (Fix( $\emptyset$ )-preservation as intertwining)."""
    return np.allclose(T @ Phi_src, Phi_tgt @ T, atol=tol)

# -----
# Modality: gauge-invariant projector
# -----
def gauge_projector(reps: List[np.ndarray]) -> np.ndarray:
    """ $P_G = (1/|G|) \sum_g \rho(g)$ ; idempotent if reps form a representation."""
    k = len(reps)
    P = sum(reps) / k
    assert np.allclose(P @ P, P, atol=1e-8), \
        "P_G is not idempotent – input is not a representation."
    return P

# -----
# Spectral-dimension estimator via heat-trace small-t slope
# -----
def spectral_dimension(eigenvalues: np.ndarray,
                     t_small: float = 1e-4,
                     t_large: float = 1e-1) -> float:
    """Estimate  $d_{\text{sp}} = -2 \cdot d \log P(t) / d \log t$ ."""
    def P(t):
        return float(np.sum(np.exp(-t * eigenvalues ** 2)))
    return -2 * (np.log(P(t_small)) - np.log(P(t_large))) \
        / (np.log(t_small) - np.log(t_large))

# -----
# Self-test
# -----
def demo_sigma21():
    names = [f" $\sigma_{\{i\}}$ " for i in range(21)]
    N = 21
    D = np.zeros((N, N))
    for i in range(1, N):
        D[i, i - 1] = 1 # simple chain  $\sigma_0 \rightarrow \sigma_1 \rightarrow \dots \rightarrow \sigma_{\{N-1\}}$ 
    E = np.zeros(N)
    E[0] = 1 #  $\sigma_0$  is empirically activated
    reg = SigmaRegistry(names, D, E)
    fp, iters, resid = reg.phi_fixed_point()
    print(f"Iters to fixed point: {iters}")
    print(f"Residual: {resid:.3e}")
    print(f"Fixed point: {fp.astype(int)}")
    return reg, fp, resid

if __name__ == "__main__":
    demo_sigma21()

```

# Part V. Self-decoding encoding with Merkle commitment

## V.1 Goals

The encoding must (i) survive ordinary text transmission (email, chat, web paste, copy-paste between editors); (ii) carry a hash commitment to its own specification, so a recipient can verify that the payload's decoder bytes match the hash; (iii) be implementable in a few dozen lines of code in any modern language.

**Editor's note — reframed from "Löb closure".** The original manuscript framed the encoding as a "Löb-style fixed point". On review, what the construction actually is is a Merkle-style hash commitment:  $\text{payload} = \text{spec} \parallel \text{H}(\text{spec}) \parallel \text{E}(\text{spec} \parallel \text{H}(\text{spec}))$ , where E is the encoder. This is a commitment scheme, not a fixed-point of an endofunctor on a modal category. The reframing is honest and the construction is identical.

## V.2 Channel: invisible Unicode variation selectors

We use the Unicode variation selectors U+FE00 through U+FE0F (VS1-VS16, sixteen codepoints, four bits each). They are invisible in nearly all renderers and survive most transmission channels intact. Reference: Unicode Standard 16.0.0 (Unicode Consortium 2024); UTR #39, Unicode Security Mechanisms.

Pair two variation selectors per byte (high nibble first, low nibble second). Append the pairs after a benign carrier character. Decoders normalise the input to NFC, scan for VS codepoints, and group in pairs.

## V.3 Information-theoretic guarantee

A steganographic scheme is  $\epsilon$ -secure if the KL divergence between the stego distribution and the cover distribution is  $\leq \epsilon$  (Cachin 2004, *Inf. Comput.* 192, 41). Our variation-selector channel has  $\text{KL} \approx 0$  with respect to ordinary text in channels that do not specifically scan for VS codepoints. Aggressive NFKC normalisation can strip VS sequences; for those channels, fall back to a homoglyph encoding via UTR #39 confusables (Gabrilovich & Gontmakher 2002, *CACM* 45(2), 128) at one bit per carrier letter.

## V.4 Reference Python implementation

```
import hashlib
import unicodedata

# Variation Selectors 1–16: U+FE00 .. U+FE0F (4 bits each)
VS_BASE = 0xFE00
VS_COUNT = 16
```

```

def byte_to_vs_pair(b: int) -> str:
    """Encode one byte as two variation selectors (hi nibble + lo nibble)."""
    hi, lo = (b >> 4) & 0xF, b & 0xF
    return chr(VS_BASE + hi) + chr(VS_BASE + lo)

def encode(payload: bytes, carrier: str = "$") -> str:
    """Embed payload bytes after the first carrier character via VS sequences."""
    if not carrier:
        carrier = "$"
    vs = "".join(byte_to_vs_pair(b) for b in payload)
    return carrier[0] + vs + carrier[1:]

def decode(stego: str) -> bytes:
    """Extract VS-encoded payload; ignores non-VS characters."""
    nfc = unicodedata.normalize("NFC", stego)
    vs_codes = [c for c in nfc
                 if VS_BASE <= ord(c) < VS_BASE + VS_COUNT]
    if len(vs_codes) % 2 != 0:
        raise ValueError("Odd number of VS codepoints; payload corrupted.")
    out = bytearray()
    for i in range(0, len(vs_codes), 2):
        hi = ord(vs_codes[i]) - VS_BASE
        lo = ord(vs_codes[i + 1]) - VS_BASE
        out.append((hi << 4) | lo)
    return bytes(out)

def merkle_payload(spec: bytes, budget: int = 4096) -> bytes:
    """Construct payload = spec || SHA256(spec) || encode(spec || SHA256(spec))."""
    h = hashlib.sha256(spec).digest()
    core = spec + h
    encoded_core = encode(core).encode("utf-8")
    payload = core + encoded_core
    return payload[:budget]

def verify(stego: str, spec: bytes) -> bool:
    """Decode, verify spec prefix and SHA-256 commitment."""
    decoded = decode(stego)
    n_spec = len(spec)
    if not decoded.startswith(spec):
        return False
    if decoded[n_spec : n_spec + 32] != hashlib.sha256(spec).digest():
        return False
    return True

if __name__ == "__main__":
    SPEC = (
        b"sigma-LANG vN. Encoding: Unicode VS U+FE00..U+FE0F nibble pairs. "
        b"Decoder: extract VS codes, group in pairs, reconstruct bytes. "
        b"Self-decoding: payload = spec || SHA256(spec) || encode(spec||hash). "
    )
    payload = merkle_payload(SPEC, budget=4096)
    stego = encode(payload, carrier="Hello – ordinary text.")
    print("Stego text (VS invisible in most renderers):")
    print(stego[:80] + "...")

```



```
print(f"Recovered {len(decode(stego))} bytes; verify: {verify(stego, SPEC)}")
```

# Part VI. Status table, novelty assessment, open problems

## VI.1 Status of every nontrivial result

| Result                                                                                          | Status                            | Evidence / proof location                                    |
|-------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------|
| $\text{Site}_T$ is a Grothendieck site (I.1.2)                                                  | PROVEN                            | Elementary; proved in §I.1.                                  |
| $\text{Sh}_\infty(\text{Site}_T)$ is a presentable $\infty$ -topos (I.1.3)                      | CONSTRUCTED                       | Lurie HTT 6.1.0.6.                                           |
| $\sigma_{21}$ : $\Phi$ -closure has least fixed point reached in $\leq  \text{registry} $ steps | PROVEN                            | Tarski 1955 + finite lattice; §I.3.                          |
| $\sigma_{73}, \sigma_{74}$ : HBL1, HBL2, HBL3 in §2 runtime                                     | PROVEN                            | §I.4.2, §I.4.3.                                              |
| $\sigma_{75}$ : Löb derived inside §2                                                           | PROVEN                            | §I.4.4 (HBL 13-step template).                               |
| Gödel 2 ( $\neg \Box \perp$ ) as stable $\Phi$ -invariant                                       | PROVEN                            | §I.4.5; consequence of $\sigma_{75}$ .                       |
| Connes spectral triple for arbitrary §-dialect                                                  | OPEN                              | Restated as Open Question I.6.1.                             |
| Structural redundancy ratio interpretation                                                      | CONSTRUCTED                       | §I.7.2; structural, not Shannon.                             |
| II.1.1 superposition under (H1), (H2)                                                           | PROVEN under hypotheses           | §II.1; truncation tower argument.                            |
| §§ functoriality (II.3.1)                                                                       | PROVEN                            | Composition of presheaf assignment with sheafification.      |
| Guarded self-reference of §§ (II.3.2)                                                           | PROVEN within guarded type theory | Gratzer 2025.                                                |
| Modalities M1–M4, M6, M8, M9, M12                                                               | PROVEN existence as fixed points  | §II.2 table.                                                 |
| Modality M5 (bifurcation), M11 (braiding)                                                       | CONSTRUCTED                       | §II.2; not strict fixed points; alternative semantics given. |
| Modality M7 (RG flow lift)                                                                      | SCAFFOLDING                       | §III.4.2; field content not specified.                       |
| Modality M10 (phase transition, RSB cases)                                                      | CONSTRUCTED                       | §II.2; ultrametric extension noted.                          |
| Modality M13 (self-reference)                                                                   | PROVEN within guarded type theory | §II.2; Gratzer 2025.                                         |
| §-hyp-n-d reduction at signature (2, −1, 2)                                                     | PROVEN                            | §III.3.1.                                                    |

| Result                                                                          | Status      | Evidence / proof location                   |
|---------------------------------------------------------------------------------|-------------|---------------------------------------------|
| $\sigma_{55}$ : $q \rightarrow 1$ classical limit recovers undeformed substrate | PROVEN      | §III.5.2; Kashaev 1998 / Chekhov–Fock 1999. |
| $N = 1$ reduction to single-factor 3-d hyperbolic                               | PROVEN      | §III.5.1; tautological.                     |
| Mixed-curvature resolution of fiber-rank collapse                               | CONJECTURED | §III.6.1; mechanism sketched, not verified. |
| Matricial Boolean realisation of $\Phi$                                         | PROVEN      | §IV.4; explicit Neumann-series form.        |
| Möbius / $SO(n,1)^+$ matricial realisation                                      | PROVEN      | §IV.2, §IV.3; standard.                     |
| Self-decoding Merkle-committed encoding scheme                                  | PROVEN      | §V.4; reference implementation provided.    |

## VI.2 Genuinely novel content

To the best of our knowledge, the following are original to this thesis (with citation gaps flagged):

**(N1)** The  $(2,1)$ -/stack-/ $\infty$ -topos triple-superposition theorem (II.1.1) for a typed-dialect family does not appear in the literature in this specific form. The ingredients (truncation tower of an  $\infty$ -topos, comparison of stacks with 1-categorical reflections) are standard.

**(N2)** The  $\sigma$ -flag registry as a finite distributive lattice with an explicit  $\sigma$ -closure operator and a  $\sigma$ -rule self-naming scheme that supplies the diagonal lemma needed for  $\sigma_{75}$  (I.4.4) is, as a concrete formalism, original to this framework. The components (Knaster-Tarski, computably-enumerated registries, Hilbert-Bernays template) are standard.

**(N3)** The  $\S$ -hyperbolic- $n$ -d signature  $(n, \kappa, d_{sp})$  with an explicit spectral-dimension axis as a  $\S\S$ -morphism-supporting parameter is, in this combination, not in the mixed-curvature ML literature. Mixed curvature alone is well-studied (Gu et al. 2019; Skopek et al. 2020; Bachmann et al. 2020); the spectral-dimension axis adds a degree of freedom not previously combined with  $\kappa$  in a learning-theoretic context.

**(N4)** The variation-selector self-decoding encoding with Merkle commitment in the specific `spec || hash || encode(spec || hash)` form (Part V) is original as a specific construction. The components (invisible-Unicode steganography, Cachin security model, hash commitment) are standard.

**(N5)** The Connes-spectral-triple correspondence is restated as an open question (I.6.1). We do not claim it as novel; we claim that the question of *which*  $\S$ -LANG dialects support a non-degenerate spectral triple is well-posed and open.

## VI.3 Open problems

**OP1.** Construct a non-degenerate Connes spectral triple for the QuantumHoloSurface dialect (or a successor whose underlying tessellation is non-amenable). Compute

summability dimension and verify all Connes regularity axioms.

**OP2.** Promote III.4.2 from SCAFFOLDING to CONSTRUCTED by specifying field content, RG scheme, and explicit identification of  $(\kappa, d_{sp})$  with running couplings.

**OP3.** Empirically and theoretically verify Conjecture III.6.1 (mixed-curvature resolution of fiber-rank collapse). Map the  $(\kappa_+, \text{training schedule})$  phase diagram.

**OP4.** Develop the RSB extension of M10 to handle genuinely ultrametric Gibbs-measure landscapes (Charbonneau et al. 2023, *Spin Glass Theory and Far Beyond*, World Scientific).

**OP5.** Determine whether  $\mathbb{S}_c$  is universal among  $\infty$ -topoi with a sufficiently rich modality structure (a  $\mathbb{S}$ -analogue of Lurie's classification of  $\infty$ -topoi as left exact localisations of presheaf  $\infty$ -categories).

**OP6.** Determine whether the auto-abstraction principle (II.3.2) gives a usable response to the tiling-agents obstacle in self-modifying decision theory. Preliminary alignment with Ahrenbach (2024, arXiv:2408.09590) and Garrabrant et al. (2016, arXiv:1609.03543) is suggestive but unresolved.

## VI.4 Closing remarks

This thesis does not claim that the framework solves the Löbian obstacle, delivers a complete theory of self-referential AI safety, or unifies topos theory with mixed-curvature learning. It claims, more modestly: that a typed-dialect family over a presheaf substrate admits a rigorous categorical foundation at  $\infty$ -topos level; that its two iterated extensions ( $\mathbb{S}$  and  $\mathbb{S}$ -hyp-n-d) are coherent under the truncation-tower superposition under explicit hypotheses; that the entire transformation algebra admits an executable matricial realisation; and that a self-decoding Merkle-committed encoding can be specified in a few dozen lines.

The framework's mathematical weight rests on  $\sigma_{21}$ ,  $\sigma_{73-75}$ , and the matricial realisation as genuine theorems and constructions. The Connes-spectral-triple correspondence and the mixed-curvature resolution of fiber-rank collapse are open questions of independent interest. Other items are scaffolding awaiting further work. The peer-review process incorporated here removed eight specific overclaims and we expect future review to find more.

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